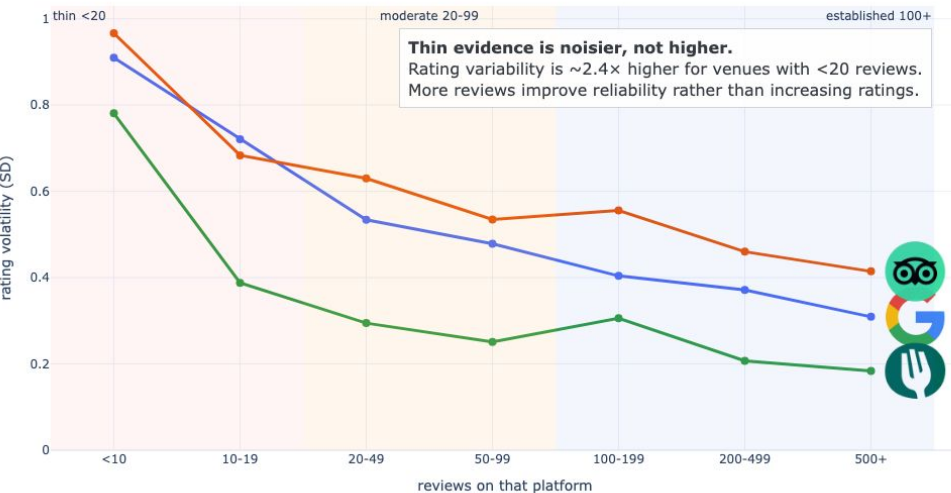


Data Visualization Pitch

Student Individual Assignment



Can we trust restaurant ratings, or does the platform we use change our perception of quality?

A visual analysis of restaurant ratings in Milan based on an integrated dataset from Google Maps, Tripadvisor, and TheFork

Online review platforms strongly influence how people perceive restaurant quality, but their ratings are not always consistent. Using an integrated dataset of more than 10,000 restaurants in Milan, this presentation explores rating consistency, platform bias, review volume, geographic patterns, and other factors that shape our perception of restaurant quality through a series of data visualizations

Student name: **Vladyslav Yelisieiev**

Student ID: **947134**

Student e-mail address: **v.yelisieiev@campus.unimib.it**

Research questions

- Do review platforms generally agree on restaurant quality?
- If not, where do the largest disagreements occur?
- What factors explain these differences?
 - Review volume
 - Platform bias
 - Restaurant characteristics (location, price and cuisine)
- What do these findings mean for users choosing a restaurant?

About Data

Dataset

The dataset was collected as part of the **Data Management** module group project ([repo](#)) of this course and is used for all analytics. It integrates restaurant information from **Google Maps, Tripadvisor, and TheFork**, resulting in **10,054 restaurants** from Milan ([link](#)). The dataset reflects the state of the source platforms as of May 2026.

Data Sources & Usage

The platforms were selected because they provide complementary restaurant information and different user communities. The data was collected from publicly available sources for educational and research purposes.

Data Quality

The main challenges included heterogeneous schemas, missing values, inconsistent naming and addresses, different platform coverage, and the absence of a common restaurant identifier. Tripadvisor records also required geocoding due to missing coordinates.

Preparation

Before visualization, the data was cleaned, standardized, matched across platforms using entity resolution, and integrated into a unified analytical dataset suitable for comparison.

Methodology

Data Collection

Restaurant data was collected from Google Places API, Tripadvisor (web scraping), and TheFork (web scraping), providing complementary information on ratings, reviews, cuisine, prices, and location.

Data Processing & Transformation

The data was processed using Python and stored in databases: MongoDB and ClickHouse. Source-specific schemas were standardized, missing values handled where possible, addresses geocoded, and restaurants matched across platforms using entity resolution to create a unified analytical dataset.

Analysis

The visual analysis combines comparative analysis, correlation analysis, and spatial analysis to investigate rating consistency, platform bias, review volume, cuisine differences, and geographic patterns. *Python* and the *Plotly* library were used for visualization.

AI Tools & Reproducibility

Claude Code and Codex were used to assist with code refinement and documentation, as well as to support the manual review of uncertain entity-matching cases. The complete data collection and processing pipeline is fully documented and reproducible through the [project repository](#).

Insights from the Data

The exploratory analysis revealed several interesting patterns that motivated a deeper visual investigation:

- **Most restaurants receive similar ratings across review platforms**, suggesting an overall level of agreement.
- **A subset of restaurants shows substantial rating differences**, raising questions about the factors behind these inconsistencies.
- **Review volume and platform coverage vary considerably**, potentially influencing the reliability of ratings.
- **Restaurant characteristics**, such as cuisine type and geographic location, may contribute to differences in ratings and data completeness.

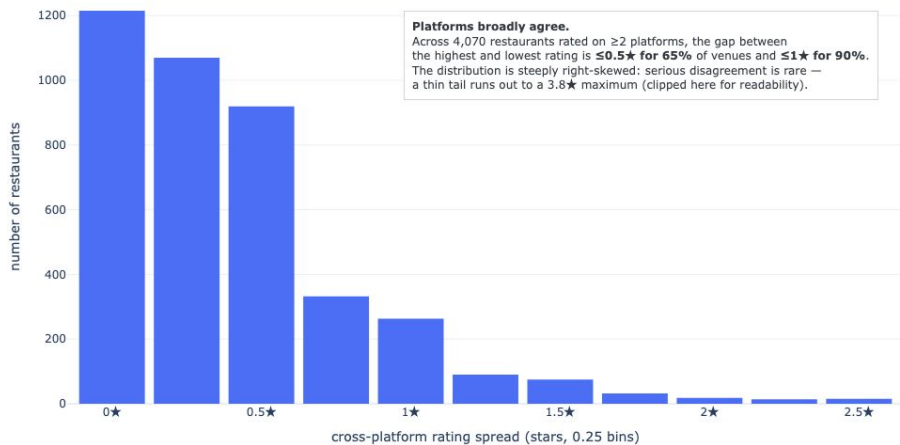
These observations motivated a series of visual analyses to better understand **rating consistency, platform bias, review reliability, and the influence of restaurant characteristics**.

Chapter 1 - How Consistent Are Restaurant Ratings Across Platforms?

In this chapter, we investigate:

- whether Google Maps, Tripadvisor, and TheFork provide consistent ratings;
- which platform contributes most to disagreement;
- which factors cause the large differences in ratings;
- whether the platforms exhibit systematic rating bias.

Do Review Platforms Agree on Restaurant Ratings?



"Cross-platform restaurant rating spread" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

Platforms Mostly Agree on Restaurant Quality

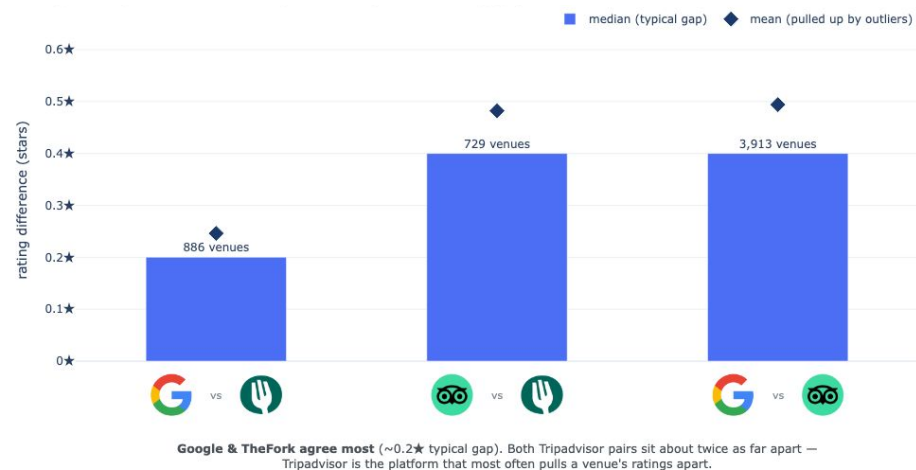
The analysis begins by measuring the difference between the highest and lowest rating assigned to the same restaurant across platforms.

The results show a **high level of consistency**:

- **65%** of restaurants differ by **no more than 0.5 $\star$**
- **90%** differ by **1 $\star$ or less**

However, a small group of restaurants exhibits substantially larger differences, suggesting that the platform used can influence how restaurant quality is perceived. **These outliers motivate the next step of the analysis.**

Which platform contributes most to disagreement?



"Pairwise disagreement rate of online platforms" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

The pairwise comparison shows that disagreement is not equally distributed across platforms.

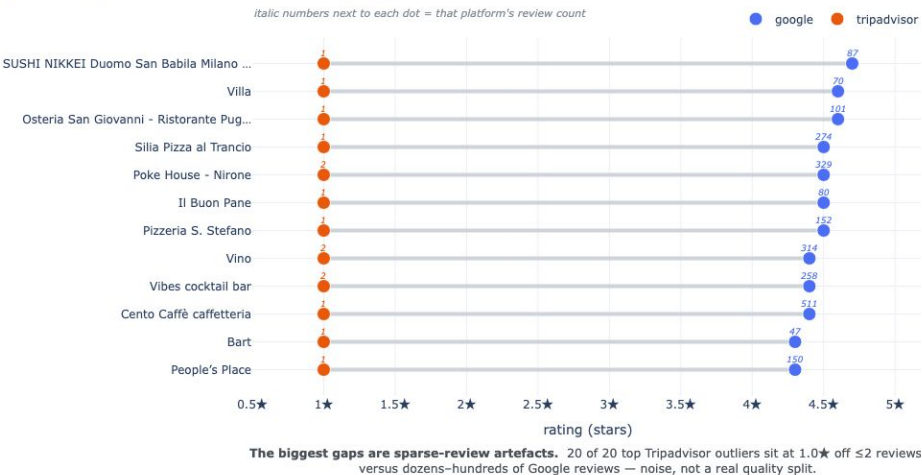
Google Maps and TheFork are the most aligned, with a typical rating gap of only around **0.2★**.

In contrast, the pairs involving **Trippadvisior** show larger gaps. Both **Trippadvisior-Google** and **Trippadvisior-TheFork** have a typical difference of around **0.4★**, and their means are pulled higher by outliers.

This suggests that **Trippadvisior is the platform that most often drives cross-platform rating disagreement**, while Google Maps and TheFork tend to provide more similar evaluations.

If Trippadvisior contributes more to disagreement, the next question is: **which restaurants are most affected by these rating differences?**

Are the Largest Disagreements Real?



"Google vs Tripadvisor: biggest rating gaps" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

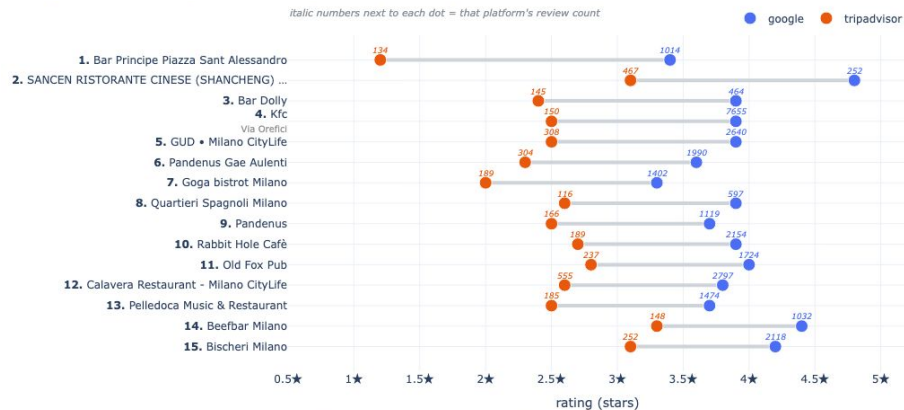
The restaurants with the **largest rating differences** between Google Maps and Tripadvisor are predominantly associated with **very few Tripadvisor reviews**, often only **one or two**. With such limited feedback, a single negative review can disproportionately lower the overall rating.

In contrast, the same restaurants typically have **dozens or even hundreds of Google reviews**, resulting in a much more stable and representative average rating.

These findings suggest that the most extreme disagreements are often **artifacts of sparse review data** rather than genuine differences in restaurant quality.

But do significant disagreements still exist when both platforms have a large number of reviews?

Do Rating Differences Persist When Both Platforms Have Sufficient Reviews?



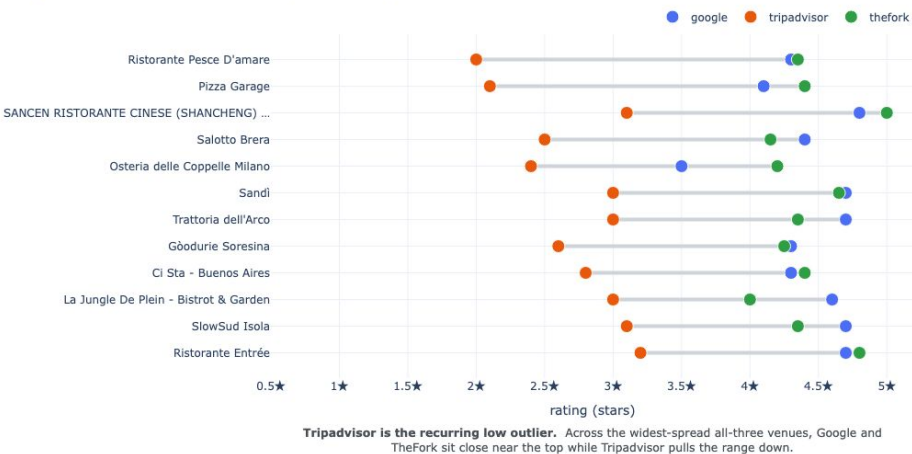
"Google vs Tripadvisor: robust disagreements" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

We established that many extreme disagreements are associated with very few Tripadvisor reviews. However, this visualization demonstrates that **substantial rating differences still exist even when both Google Maps and Tripadvisor have at least 100 reviews.**

For these restaurants, Google consistently assigns **higher ratings** than Tripadvisor, indicating that the observed differences are **unlikely to be caused by sparse data alone.** Instead, they suggest the presence of **systematic platform-specific differences** in how users evaluate restaurants.

If review volume does not fully explain these discrepancies, do similar patterns appear when comparing all three platforms?

How Do the Three Platforms Compare in the Largest Disagreement Cases?



"All-three venues: widest spread across platforms" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

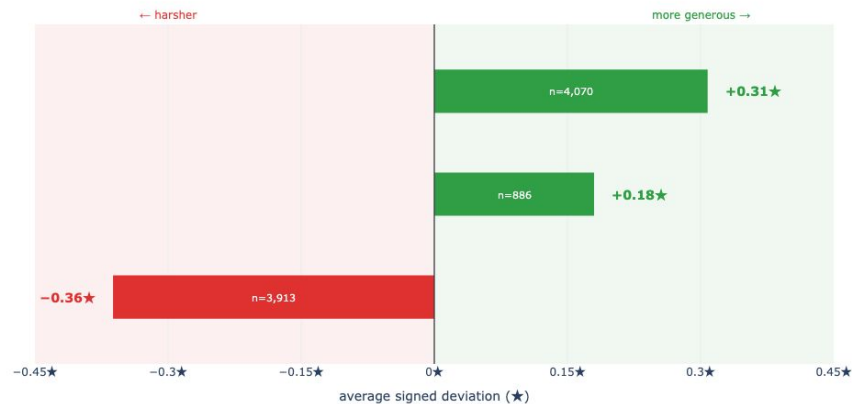
The comparison of restaurants available on **all three platforms** is consistent with the pattern observed in the previous analyses.

In the largest disagreement cases, **Google Maps and TheFork generally report similar ratings**, while **Tripadvisor consistently assigns lower ratings**. This suggests that Tripadvisor is the main contributor to cross-platform disagreement, whereas Google Maps and TheFork provide more consistent evaluations.

These findings reinforce the conclusion that **not all rating differences can be explained by sparse review data alone**. Platform-specific characteristics also appear to influence how restaurant quality is perceived.

Is Tripadvisor simply different in extreme cases, or is there a systematic platform bias?

Is There a Systematic Platform Bias?



LOO deviation = platform rating minus the average of the other platforms for the same restaurant, then averaged.
Star labels show the signed bias; n labels show rated restaurants behind each bar.

"Leave-one-out platform bias (accounting for peer consensus)" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

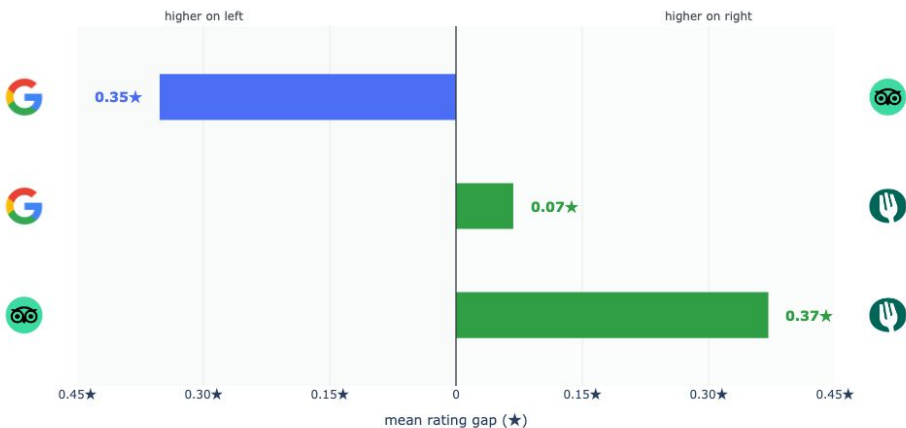
Each platform is compared with the **average rating of the other two platforms** for the same restaurant. A positive value means the platform tends to rate restaurants **higher than the consensus**, while a negative value indicates a **more conservative** tendency.

The analysis reveals a clear systematic difference. **Google Maps (+0.31★)** and **TheFork (+0.18★)** generally rate restaurants above the consensus, whereas **Tripadvisor (-0.36★)** consistently rates them below it.

This suggests that the disagreement observed throughout the previous analyses is not random but reflects a consistent difference in how the platforms evaluate restaurants.

Does this conclusion also hold when the platforms are compared directly, without using the consensus measure?

Do Pairwise Comparisons Confirm This Bias?



Bars point toward the higher-rated side, so direction carries the sign.
Tripadvisor is lower in both head-to-head checks (0.35★ and 0.37★); the Google-TheFork gap is only 0.07★
meaning that Google and TheFork pretty much align, unlike Tripadvisor.

"Pairwise average rating gap (★)" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

Each bar represents the **average rating difference between two platforms**. The bar points toward the platform with the **higher average rating**, while its length indicates the magnitude of the difference.

The pairwise comparison is consistent with the previous findings. **Google Maps and TheFork differ by only 0.07★**, indicating a high level of agreement. In contrast, both platforms rate restaurants approximately **0.35-0.37★ higher than Tripadvisor**, confirming that Tripadvisor is systematically more conservative rather than producing occasional outliers.

Chapter 1 - Summary

Key Findings

- Restaurant ratings are **generally consistent** across platforms.
- **Google Maps and TheFork** show the highest level of agreement.
- **Tripadvisor** contributes most to cross-platform disagreement.
- Many extreme differences are explained by **very few Tripadvisor reviews**, although some disagreements remain even when both platforms have abundant review data.
- Tripadvisor consistently assigns lower ratings than Google Maps and TheFork, making it the primary source of **bias** in cross-platform disagreement.

Conclusion

Restaurant ratings can generally be trusted, but users should consider both the review platform and the number of available reviews when interpreting rating differences. Tripadvisor tends to be more conservative than Google Maps and TheFork.

Chapter 2 – Beyond Rating Consistency: What Shapes Restaurant Ratings?

In this chapter, we investigate:

- whether review volume affects the stability and reliability of restaurant ratings;
- how location within Milan influences the completeness of restaurant information;
- whether cuisine type and price range are associated with differences in restaurant ratings;
- whether platform visibility reflects restaurant quality or simply greater popularity.

Does Review Volume Influence Rating Reliability?



"More reviews → less volatile ratings" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

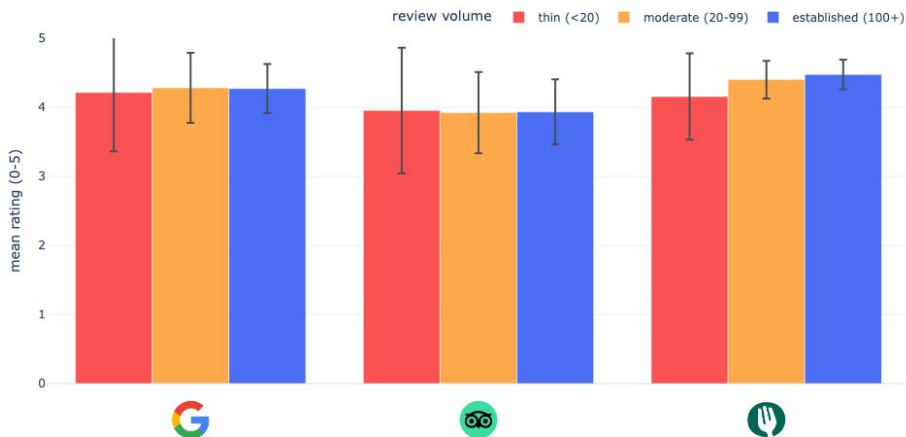
Restaurants are grouped by the number of reviews they receive on each platform. The y-axis shows the **standard deviation of ratings** within each group, where higher values indicate greater variability and lower rating stability.

Across all three platforms, rating volatility **decreases as the number of reviews increases**. Restaurants with fewer than **20 reviews** show substantially more variable ratings, while establishments with over **100 reviews** exhibit much more stable evaluations.

This suggests that **review count is a strong indicator of rating reliability**. A high rating based on only a few reviews should therefore be interpreted with greater caution.

Does a small number of reviews actually increase the average rating, or does it simply make ratings more uncertain?

Do Restaurants with Few Reviews Receive Higher Ratings?



No inflation at the mean. The three tier means sit within $\sim 0.32\star$ on every platform; where they differ at all, thin venues are rated slightly lower, not higher (Google thin vs established mean-diff $p = 0.02$). The 'sparse looks better' intuition does not hold.

"The rating mean barely moves; only the spread shrinks" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

Restaurants are divided into **thin** (<20 reviews), **moderate** (20-99), and **established** (100+) groups. The bars show the **average rating**, while the whiskers represent **± 1 standard deviation**, illustrating the spread of ratings within each group.

The average rating changes very little across review-volume groups. Instead, the main difference lies in the **variability** of ratings: restaurants with **few reviews show much wider distributions**, while highly reviewed restaurants have much more consistent evaluations.

This indicates that **low review counts do not systematically inflate ratings**, they simply make them **less reliable**.

If the average remains stable, what do these more variable rating distributions actually look like?

How Does Review Volume Affect Rating Distribution?



Same average, very different risk. On Google 35% of thin venues sit at ≥ 4.8 ★ (a 3-review 5.0) vs just 6% of established ones — with a matching low tail. A high rating on a thin venue is unreliable in both directions.

"Thin-evidence venues polarize (5★ pile-up + a low tail)" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

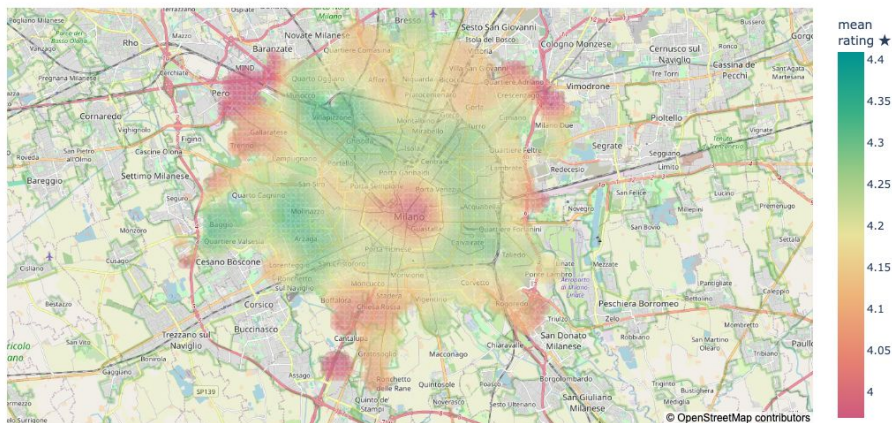
The violin plots compare the distribution of ratings for restaurants with **few reviews (<20)** and **many reviews (100+)**. Wider sections represent rating values that occur more frequently.

Restaurants with very few reviews exhibit a **much broader and more polarized distribution**. While many receive perfect or near-perfect ratings, they also show a longer tail of very low ratings. In contrast, restaurants with many reviews are concentrated around a stable average with far less variability.

These findings reinforce that **review volume affects the reliability of ratings rather than their average value**.

But are highly rated restaurants distributed uniformly across Milan, or do some areas consistently achieve better ratings?

Do Restaurant Ratings Vary Across Milan?



The tourist core rates lower. The historic centre averages only ~4.00★ (red), while ratings peak in the 2-4 km belt (Navigli, Isola, Porta Romana) at ~4.24★ (green) before easing off in the far periphery.

"Distribution of restaurant ratings across Milan" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

The map shows the **average restaurant rating** for approximately 1 km grid cells across Milan. Warmer colors represent higher average ratings, while cooler colors indicate lower-rated areas.

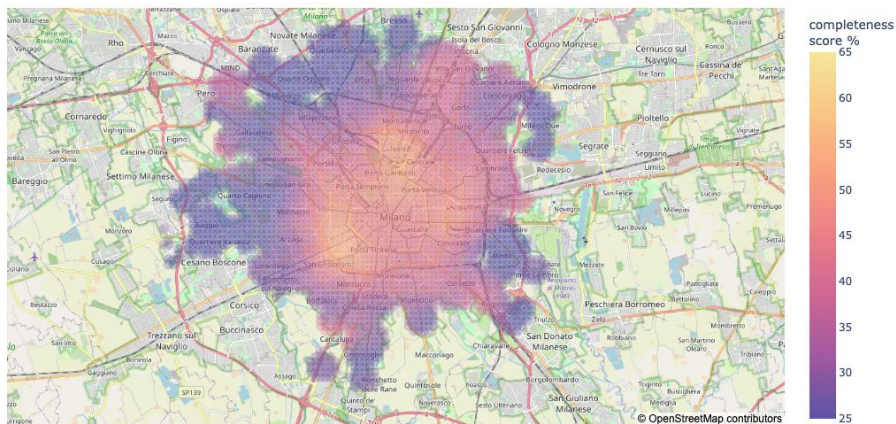
Restaurant ratings are **not uniformly distributed across the city**. Interestingly, the **historic city center** has slightly lower average ratings (around 4.0★), while the **2-4 km belt**, including **Navigli, Isola, and Porta Romana**, contains the highest-rated restaurants, with average ratings reaching approximately 4.24★.

Beyond this ring, ratings gradually decrease again toward the outer districts.

These results suggest that **restaurant quality is influenced by location**, but the relationship is more nuanced than a simple "city center is best" assumption.

If restaurant ratings vary geographically, is the same true for the amount of information available about those restaurants?

Does Restaurant Location Affect Data Completeness?



The centre is data-rich. Completeness is higher in the historic core (~47%), peaks in Navigli and Isola neighbourhoods, and fades smoothly outward to ~30% in the outer ones (Niguarda, Comasina etc) — a gradient along popular streets and areas.

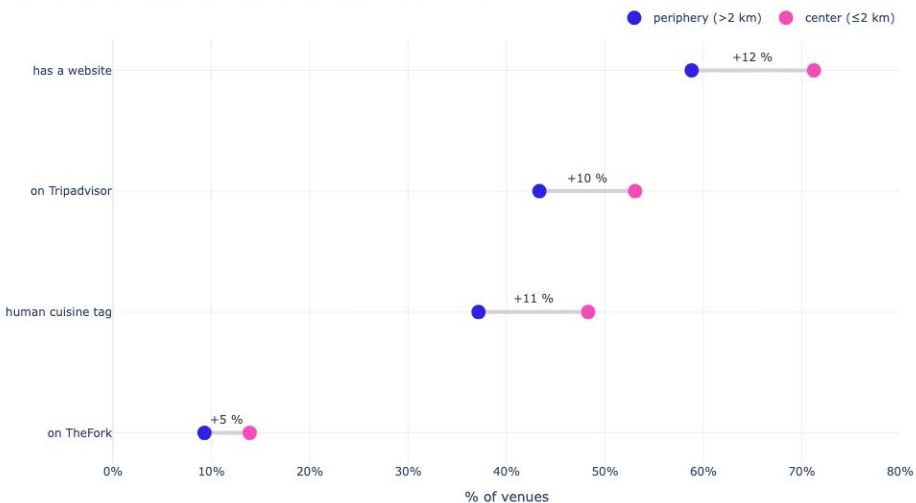
"Distribution of profile completeness across Milan" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

The map shows the **average completeness score** of restaurants across Milan. The score combines four information sources available for each restaurant (**website, cuisine tag, Tripadvisor listing, and TheFork listing**). Higher values indicate more complete restaurant profiles.

The highest data completeness is concentrated in **central Milan**, particularly around the **Duomo, Brera, Navigli, and Isola**, where restaurant profiles contain the richest online information. Moving away from the historic center, completeness gradually decreases. Outer districts such as **Niguarda, Comasina, Baggio, and Quarto Oggiaro** exhibit noticeably lower completeness scores, indicating that restaurants in peripheral areas tend to have less comprehensive online profiles. Rather than a sharp boundary, the visualization reveals a **smooth center-to-periphery gradient**, with completeness decreasing continuously as the distance from the city center increases.

Which pieces of information are actually driving this difference between the city center and the outskirts?

Which Data Components Differ Between the Center and the Periphery?



"Central venues carry fuller data (center vs periphery)" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

The chart compares the percentage of restaurants containing each data component between **central** (≤ 2 km) and **peripheral** (> 2 km) areas.

The center-periphery pattern is consistent across all major completeness components.

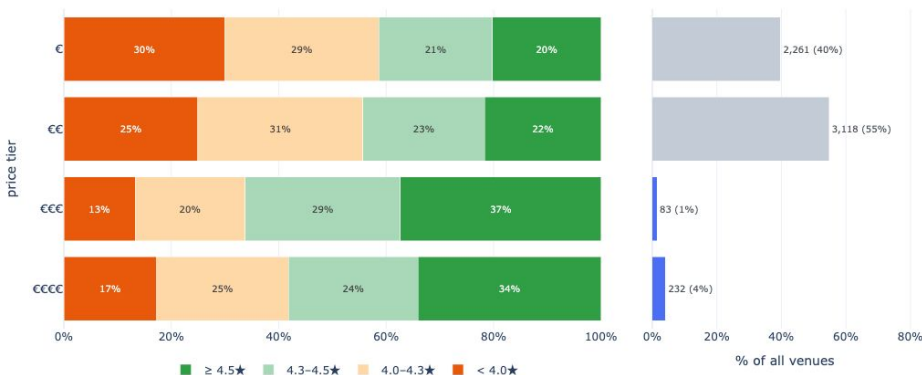
Restaurants located in central Milan are more likely to have:

- a website (+12%),
- cuisine information (+11%),
- a Tripadvisor listing (+10%),
- and a TheFork listing (+5%).

In contrast, some attributes—such as **Google review availability**, **phone numbers**, and **photos**—are almost universally available and therefore **do not contribute to location-based differences in completeness**.

Location influences the amount of available information, but do restaurant characteristics such as price also influence restaurant quality?

Does Paying More Always Mean Better Ratings?



More premium venues provide better value, but to an extent. Within the rare €€€€/€€€€ tiers the top-rated (≥4.5★) share jumps to ~37% and the weakest (<4.0★) share halves to ~13% (left). But only ~6% of Milan's venues are €€€/€€€€ (right); across the 94% at € / €€ the rating mix is essentially identical — trading up within the mainstream changes nothing.

"You get what you pay for only at the top, and the top is rare" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

The stacked bars show the distribution of ratings within each price category, while the panel on the right indicates how common each price tier is in the dataset.

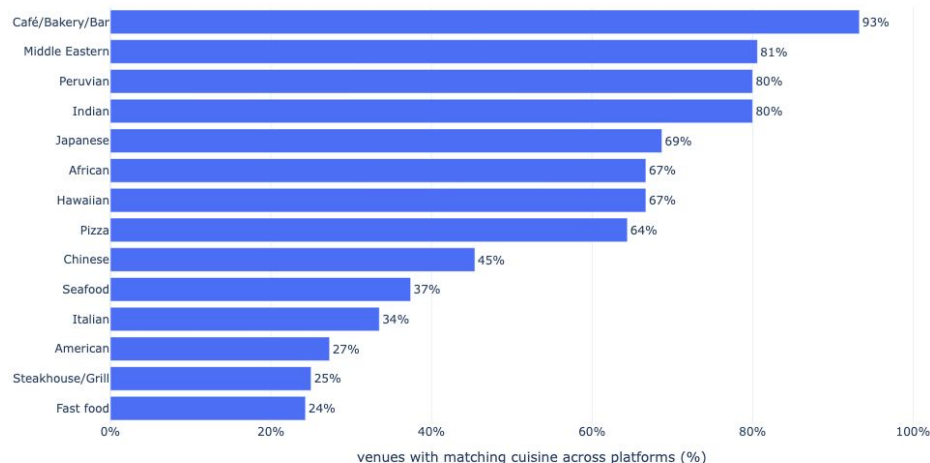
Price influences ratings **only at the premium end.**

Restaurants in the € and €€ categories display nearly identical rating distributions, indicating that spending slightly more does not improve quality.

Only the rare €€€ and €€€€ restaurants show a larger share of top-rated venues (≥4.5★ rises to ~37%), but since these tiers make up just ~6% of all restaurants, this quality premium reaches only a **tiny slice of the market.**

And do review platforms even agree on how restaurants should be classified by cuisine?

Do Review Platforms Agree on Restaurant Cuisine?



Broad categories collide; niche ones don't. For Italian, seafood and fast food the platforms agree on the cuisine barely a third of the time — one calls it Italian, another Pizza or Seafood, because the broad buckets overlap on the menu. Specific cuisines leave no room for argument: Indian, Peruvian and Middle Eastern land at 80%+. So part of "disagreement" is a data quality problem, not a taste one.

"Do the platforms agree what a place is" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

The chart shows the percentage of restaurants for which multiple platforms assign the same cuisine category.

Agreement depends strongly on the specificity of the cuisine. Distinct cuisines such as **Middle Eastern**, **Indian**, and **Peruvian** show high agreement (around **80%**), whereas broad categories such as **Italian**, **Seafood**, and **Fast Food** exhibit much lower agreement.

This is a **separate kind of disagreement**: the platforms differ on how a restaurant is categorized, not on how good it is. It's a **labeling inconsistency**, and it matters for any cuisine-based comparison — like the ones that follow.

Do restaurant characteristics, such as cuisine type, also influence how restaurants are represented and evaluated?

Do Some Cuisines Receive Better Ratings Than Others?



"Rating level vs cross-platform disagreement, by cuisine" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

Each bubble represents a cuisine. The x-axis shows the **average rating**, the y-axis the **average disagreement between platforms**, and the bubble size indicates the number of restaurants.

Cuisine influences **both restaurant ratings and cross-platform agreement**.

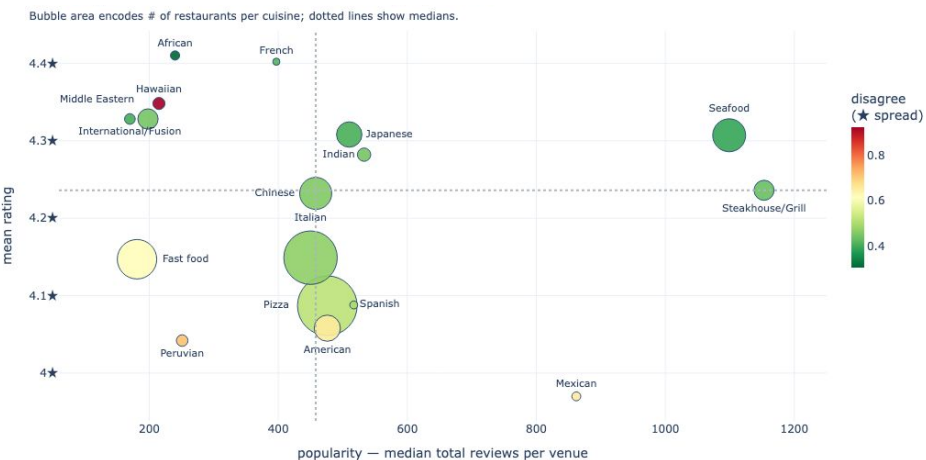
Broad, high-volume cuisines such as **Pizza**, **Fast Food**, and **American** tend to receive **lower ratings** while also exhibiting the **largest disagreement** across platforms.

In contrast, cuisines such as **Seafood**, **Japanese**, **French**, and **African** are both **highly rated** and **consistently evaluated**, suggesting stronger agreement on restaurant quality.

The most notable outlier is **Hawaiian cuisine**, which receives very high ratings but also the highest disagreement between platforms.

Does popularity explain these differences, or can less popular cuisines still achieve the highest ratings?

Does Popularity Lead to Higher Ratings?



Crowds don't track quality. Where a cuisine sits left-right (how busy) barely predicts how high it sits (how well-rated):
busy & loved — seafood, steakhouse · busy but middling — Mexican · quiet gems — African, French, Middle Eastern

"Popular ≠ loved: review volume vs rating, by cuisine" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, licensed under CC BY 4.0

This chart asks whether the crowd's attention tracks quality: each cuisine is placed by how many reviews it typically draws (left to right) against how well it's rated (bottom to top), with bigger bubbles for cuisines covering more restaurants and warmer colors where the platforms disagree most.

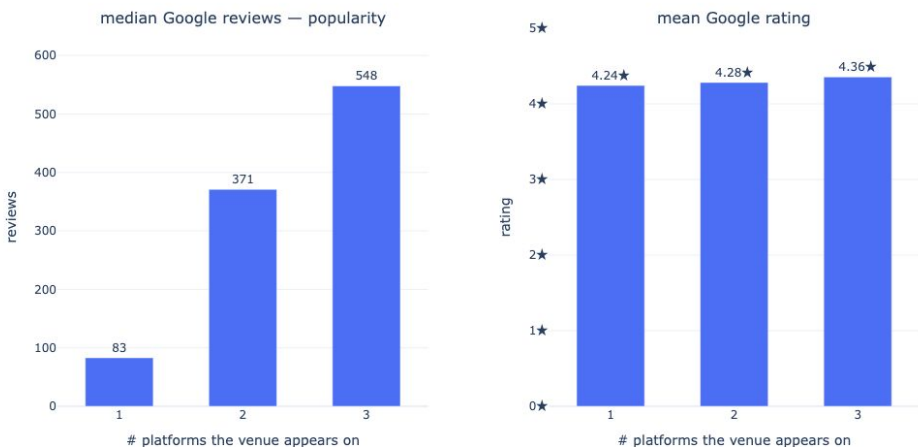
Popularity alone is **not a reliable predictor of restaurant quality**.

Some highly reviewed cuisines, such as **Seafood**, combine popularity with excellent ratings, whereas others, such as **Mexican**, receive many reviews but comparatively lower ratings.

Conversely, niche cuisines—including **African**, **French**, and **Middle Eastern**—achieve some of the highest ratings despite attracting far fewer reviews.

Besides cuisine, does appearing on more platforms mean a restaurant is actually better?

Does Being Listed on More Platforms Mean Higher Quality?



It's popularity, not the score. From 1 to 3 platforms the median review count multiplies 6.6× (83→548) — the left bars rocket — while mean rating rises only +0.11★, so on a 0–5★ scale the right bars look the same. Multi-platform venues are more popular, not better rated.

"More platforms ⇒ far more reviews, but barely higher rating" by Vladyslav Yelisieiev, based on an integrated dataset of restaurant reviews collected from online platforms, Licensed under CC BY 4.0.

Google Maps is used as the reference platform due to its highest restaurant coverage in the dataset. The left panel compares the median number of Google reviews for restaurants listed on one, two, or three platforms, while the right panel compares their average Google rating.

Restaurants listed on **more platforms** attract substantially **more reviews**, indicating higher popularity and online visibility.

However, their **average rating changes very little**. Restaurants appearing on all three platforms receive only a slightly higher average rating than those listed on a single platform, despite collecting **over 6 times more reviews**.

This suggests that **platform presence is primarily a marker of popularity and visibility rather than restaurant quality**.

Chapter 2 - Summary

Key Findings

- **Review volume** improves the reliability of ratings but does not systematically increase them.
- **Central restaurants** tend to have more complete online profiles than those in the outskirts.
- **Cuisine** influences both rating level and cross-platform agreement.
- **Premium restaurants** achieve higher ratings only in the highest price tiers.
- **Greater platform visibility** is associated with popularity rather than better ratings.

Conclusion

Restaurant ratings should be interpreted together with contextual factors. Popularity, location, cuisine, and data availability all shape how restaurants are perceived, meaning that a single rating value rarely tells the whole story.

LICENCE

Slides are shared with the following license:

'Quality of Online Restaurant Reviews in the Milan Area' by Vladyslav Yelisieiev, is licensed under a **Creative Commons Attribution 4.0 International (CC BY 4.0)** license.

You are free to:

- **Share** – copy and redistribute the material in any medium or format.
- **Adapt** – remix, transform, and build upon the material for any purpose.

Under the following condition:

- **Attribution** – appropriate credit must be given to the original author.